

* Defn A set A is called a proper subset of set B if $A \subset B$ and $A \neq B$. We write $A \subset B$

* Defn $\mathbb{N}$: The set of natural numbers
$$\mathbb{N} := \{1, 2, 3, \dots\}$$

* Defn Set theoretic difference of A and B / The complement of B relative to A .
$$A \setminus B := \{x: x \in A \text{ and } x \notin B\}$$

* Defn Sets A and B are called disjoint if
$$A \cap B = \emptyset$$

* Lemma
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

* Theorem (Principle of Induction). Let $P(n)$ be a statement depending on the natural number n . Suppose that

① $P(1)$ is true

② If $P(n)$ is true then $P(n+1)$ is true

Then $P(n)$ is true for all $n \in \mathbb{N}$

Hint: Use well ordering property of $\mathbb{N}$

* Defn Well ordering property of $\mathbb{N}$. Every non empty set of $\mathbb{N}$ has a smallest element.

* Defn Let A and B be sets. The Cartesian product is the set of tuples defined by

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

domain of f codomain of f

* Defn A function $f: \overset{\text{domain of } f}{A} \rightarrow \overset{\text{codomain of } f}{B}$ is the subset f of $A \times B$ st for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x, y) \in f$. We write $f(x) = y$

$$\underset{\text{range of } f}{R(f)} := \{y \in B : \exists x \in A \text{ st } f(x) = y\}$$

* Defn $f: A \rightarrow B$. Image / Direct Image of $C \subset A$
 $f(C) := \{f(x) \in B : x \in C\}$
 Inverse Image of $D \subset B$
 $f^{-1}(D) := \{x \in A : f(x) \in D\}$

* Lemma $f: A \rightarrow B$. Let C, D be subsets of B then
 $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$
 $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$
 $f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$

* Lemma $f: A \rightarrow B$. Let C, D be subsets of A . then
 $f(C \cup D) = f(C) \cup f(D)$
 $f(C \cap D) \subset f(C) \cap f(D)$

* Defn Given a set A , a binary relation on A is a subset $R \subset A \times A$ which consists of those pairs where the relation holds. Instead of $(a, b) \in R$ we write $a R b$.

* Defn Let A and B be sets. We say A and B have the same cardinality when there exists a bijection $f: A \rightarrow B$.

* Defn If A have same cardinality as $\{1, 2, 3, \dots, n\}$ for some $n \in \mathbb{N}$, we write $|A| := n$. If A is empty we write $|A| := 0$.
 equivalence class of all sets with same cardinality as A } finite cardinality

* Defn We write $|A| \leq |B|$ if there exists an injection from A to B .

* Defn If $|A| = \aleph$ we say A is countably infinite.

* Defn If A is finite or countably infinite we say A is countable.

* Defn The power set of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

* Theorem Let A be a set. Then $|A| < |\mathcal{P}(A)|$. In particular there exists no surjection from A onto $\mathcal{P}(A)$.

Proof Hint: $S = \{x \in A : x \notin f(x)\}$

Proof by contradiction: Assume a surjection f exists.
Let $C \in A$ s.t. $f(C) = S$

Defn - An ordered set is a set S together with a relation ' $<$ ' s.t

① $\forall x, y \in S$, exactly one of the following holds, i.e.

$$x < y \text{ or } x > y \text{ or } x = y$$

② If $x, y, z \in S$ s.t $x < y$ and $y < z$ then $x < z$

Defn Let $E \subset S$ where S is an ordered set

* If $\exists b \in S$ s.t $\forall x \in E \quad x \leq b$
 b is called an upper bound of E

* If $\exists b_0 \in S$ s.t b_0 is an upper bound of E and if b is another upper bound of E then $b_0 \leq b$ then b_0 is called least upper bound or supremum of E

* If $\exists b \in S$ s.t $\forall x \in E \quad x \geq b$, then b is called a lower bound of E

* If $\exists b_0 \in S$ s.t b_0 is a lower bound of E and if b is another lower bound of E then $b \leq b_0$, then b_0 is called greatest lower bound or infimum of E

Defn An ordered set S has the least upper bound property if every non empty set $E \subset S$ that is bounded above has a least upper bound

Defn A set F is called field if it has two operations, addition (+) and multiplication, defined on it s.t

① If $x, y \in F$ then $x + y \in F$

② $\forall x, y \in F \quad x + y = y + x$

③ $\forall x, y, z \in F \quad (x + y) + z = x + (y + z)$

④ $\exists 0 \in F$ s.t $\forall x \in F \quad x + 0 = 0 + x = x$

⑤ $\forall x \in F \quad \exists -x \in F$ s.t $x + (-x) = 0$

} Abelian group under addition

⑥ If $x, y \in F$ then $x \cdot y \in F$

⑦ $\forall x, y \in F \quad x \cdot y = y \cdot x$

⑧ $\forall x, y, z \in F \quad (x \cdot y) \cdot z = x \cdot (y \cdot z)$

⑨ $\exists 1 \in F$ s.t $1 \neq 0$ and $\forall x \in F \quad 1 \cdot x = x \cdot 1 = x$

⑩ $\forall x \in F$ s.t $x \neq 0 \quad \exists x^{-1} \in F$ s.t $x \cdot x^{-1} = x^{-1} \cdot x = 1$

} Abelian group under multiplication

⑪ $\forall x, y, z \in F \quad x \cdot (y + z) = xy + xz$

* Defn An field F is called an ordered field if it is an ordered set s.t

① $\forall x, y, z \in F \quad y < z \Rightarrow x+y < x+z$

② If $x, y \in F$ s.t $x > 0$ and $y > 0$ then $xy > 0$

* Lemma Let F be an ordered field and $x, y, z, w \in F$. Then

① If $x > 0$ and $y < z$ then $xy < xz$

② If $x \neq 0$ then $x^2 > 0$ corollary 1.70 Proof Hint $-x = -1x \Rightarrow -(-1) = 1$

③ If $0 < x < y$ then $0 < \frac{1}{y} < \frac{1}{x}$

④ $0 < x < y$ then $x^2 < y^2$

⑤ If $x \leq y$ and $z \leq w$ then $x+z \leq y+w$

* Lemma Let F be an ordered field with least upper bound property. If $A \subset F$ s.t A is non empty and bounded below then $\inf(A)$ exists.

* Defn There exists a unique ordered field $\mathbb{R}$ with least upper bound property s.t $\mathbb{Q} \subset \mathbb{R}$

* Lemma If $x \in \mathbb{R}$ s.t $x \leq \varepsilon \quad \forall \varepsilon \in \mathbb{R}$ where $\varepsilon > 0$ then $x \leq 0$

* Theorem If $x, y \in \mathbb{R}$ and $x > 0$ then $\exists n \in \mathbb{N}$ s.t $nx > y$
(Archimedean Property)

Proof Hint: Rephrase $\mathbb{N} \subset \mathbb{R}$ doesn't have an upper bound
contradiction: Let $b := \sup(\mathbb{N}) \quad \exists n > b-1$ where $n \in \mathbb{N}$
not $> b$

* Theorem If $x, y \in \mathbb{R}$ and $x < y$ then $\exists$ an $z \in \mathbb{Q}$ s.t $x < z < y$
($\mathbb{Q}$ is dense in $\mathbb{R}$)

Proof Hint Let $x > 0$ then $\exists n \in \mathbb{N}$ s.t $n(y-x) > 1$ — ①

Define $A := \{k \in \mathbb{N} : k > nx\}$. $\mathbb{N}$ well ordering prop.
 $m \in A$ is the smallest element $m > nx \Rightarrow x < m/n$
 $m-1 \leq nx \Rightarrow m-1 \leq nx < m \Rightarrow m/n < y$
from ①

* Defn

Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x+A := \{ x+y \in \mathbb{R} : y \in A \}$$

$$xA := \{ xy \in \mathbb{R} : y \in A \}$$

* Lemma

If A is bounded above

$$\sup(x+A) = x + \sup(A)$$

" " " " below

$$\inf(x+A) = x + \inf(A)$$

If $x > 0$, A is bounded above

$$\sup(xA) = x \sup(A)$$

" $x > 0$, A " " below

$$\inf(xA) = x \inf(A)$$

" $x < 0$, A " " above

$$\inf(xA) = x \sup(A)$$

" $x < 0$, A " " below

$$\sup(xA) = x \inf(A)$$

* Lemma

Let $A, B \subset \mathbb{R}$ be non empty subsets s.t $x \leq y \forall x \in A$ and $y \in B$, then

$$\sup(A) \leq \inf(B)$$

* Lemma

If $S \subset \mathbb{R}$ is a non empty set, bounded above, then $\forall \epsilon > 0 \exists x \in S$ s.t

$$\sup(S) - \epsilon < x \leq \sup(S)$$

* Defn

Let $A \subset \mathbb{R}$ be a set

(i) If A is empty then

$$\sup(A) := -\infty$$

$$\inf(A) := \infty$$

(ii) If A is not bounded above then

$$\sup(A) := \infty$$

(iii)

" " " " below then

$$\inf(A) := -\infty$$

* Defn

When a set $A \subset \mathbb{R}$ is bounded above s.t $\sup(A) \in A$, then we denote $\sup(A)$ by $\max(A)$. Similarly $\min(A)$ is defined

* Defn Absolute value function $|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

* Lemma * $|x| \geq 0$ and $|x| = 0$ if and only if $x = 0$

* $|xy| = |x||y|$

* $|x| \leq y$ if and only if $-y \leq x \leq y$

* $-|x| \leq x \leq |x| \quad \forall x \in \mathbb{R}$

* Lemma * $|x+y| \leq |x|+|y| \quad \forall x, y \in \mathbb{R}$

* $|x-y| \leq |x|+|y| \quad "$

* $||x|-|y|| \leq |x-y| \quad "$

* Defn $f: D \rightarrow \mathbb{R}$. We say f is bounded if $\exists M \in \mathbb{R}$ st $|f(x)| \leq M \quad \forall x \in D$

* Lemma If $f: D \rightarrow \mathbb{R}$ and $g: D \rightarrow \mathbb{R}$ are bounded functions and $f(x) \leq g(x) \quad \forall x \in D$, then

$$\sup_{x \in D} f(x) \leq \sup_{y \in D} g(y)$$

$$\inf_{x \in D} f(x) \leq \inf_{y \in D} g(y)$$

* Defn Intervals $[a, b] = \{x \in \mathbb{R} \text{ st } a \leq x \leq b\}$
 $(a, b) = \{x \in \mathbb{R} \text{ st } a < x < b\}$

* Defn All the intervals have the same cardinality

Example $\tan: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow (-\infty, \infty)$ is a bijective map

$\therefore (-\frac{\pi}{2}, \frac{\pi}{2})$ have same cardinality as $\mathbb{R}$

* Theorem $\mathbb{R}$ is uncountable

Proof hint:- Let $X \subset \mathbb{R}$ be countably infinite s.t if $a < b$ and $a, b \in X$ then

$\exists x \in X$ s.t $a < x < b$

countably infinite $\therefore \exists$ a bijection $f: \mathbb{N} \rightarrow X$ Let $x_1 = f(1)$

$a_1 := x_1$ $b_1 := x_{d1}$

Construct

$a_1 < a_2 < a_3 \dots < a_n < b_n < b_{n+1} \dots < b_2 < b_1$

s.t $x_j \in (a_k, b_k)$ where $j=1, 2, \dots, k$

$A = \{a_1, a_2, \dots\}$

$B = \{b_1, b_2, \dots\}$

$\alpha = \sup(A) \leq \inf(B)$

Prove $\alpha \notin A$ and $\alpha \notin B$ and $\alpha \notin X$

$\therefore X$ cannot contain all elements of $\mathbb{R}$

$\therefore \mathbb{R}$ is uncountable

* Defn

For infinite sequence of digits $0.d_1d_2d_3\dots$

$$D_n = \frac{d_1}{10} + \frac{d_2}{10^2} + \dots + \frac{d_n}{10^n}$$

Lemma: Every infinite sequence of digits $0.d_1d_2\dots$ represents a unique real number $x \in [0, 1]$, s.t

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

** For every $x \in (0, 1]$ $\exists$ a unique infinite sequence of digits $0.d_1d_2d_3\dots$ that represents x s.t

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

* Defn A sequence of real numbers is a function $z: \mathbb{N} \rightarrow \mathbb{R}$.
Another notation $\{z_n\}_{n=0}^{\infty}$

* Defn A sequence $\{z_n\}_{n=1}^{\infty}$ is called bounded if the underlying function is bounded.

* Defn A sequence $\{z_n\}_{n=0}^{\infty}$ is said to converge to a $x \in \mathbb{R}$ if $\forall \varepsilon > 0, \varepsilon \in \mathbb{R} \exists M \in \mathbb{N}$
s.t. $|z_n - x| < \varepsilon \quad \forall n \geq M$.
Limit of the sequence

$$\lim_{n \rightarrow \infty} z_n = x$$

* Defn A sequence that is not convergent is called divergent

* Lemma A convergent sequence has a unique limit.

* Lemma A convergent sequence is bounded

Proof Hint: Use triangle inequality

* Defn $\{z_n\}_{n=0}^{\infty}$ is monotone increasing if $z_{n+1} \geq z_n \quad \forall n \in \mathbb{N}$
" " " decreasing if $z_{n+1} \leq z_n \quad \forall n \in \mathbb{N}$

* Theorem A monotone sequence is bounded if and only if it is convergent
if $\{z_n\}_{n=0}^{\infty}$ is monotone increasing $\lim_{n \rightarrow \infty} z_n = \sup \{z_n : n \in \mathbb{N}\}$
" " " decreasing " " = $\inf \{z_n : n \in \mathbb{N}\}$

* Lemma Let $S \subset \mathbb{R}$ be a nonempty bounded set. Then $\exists$ monotone sequences

$$\{x_n\}_{n=0}^{\infty}, \{y_n\}_{n=0}^{\infty} \text{ s.t. } x_n, y_n \in S \text{ and}$$

$$\sup(S) = \lim_{n \rightarrow \infty} x_n$$

$$\inf(S) = \lim_{n \rightarrow \infty} y_n$$

* Defn: For a sequence $\{x_n\}_{n=1}^{\infty}$, the K -tail, where $K \in \mathbb{N}$, is the sequence

$$\{x_{n+K}\}_{n=1}^{\infty} \quad \text{or} \quad \{x_n\}_{n=K+1}^{\infty}$$

* Lemma For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent;

* $\{x_n\}_{n=1}^{\infty}$ converges

* The K tail $\{x_{n+K}\}_{n=1}^{\infty}$ converges for $\forall K \in \mathbb{N}$

* " " , " " converges for some $K \in \mathbb{N}$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_{n+K}$$

* Defn let $\{x_n\}_{n=1}^{\infty}$ be a sequence. let $\{n_i\}_{i=1}^{\infty}$ be a strictly increasing sequence of natural number.

$\{x_{n_i}\}_{i=1}^{\infty}$ is called a subsequence

* Lemma If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence. Then every subsequence $\{x_{n_i}\}_{i=1}^{\infty}$ is convergent.

* Lemma